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## Software Requirements Specification (SRS) Document

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### 1. Introduction

#### 1.1 Purpose

This document defines the functional and non-functional requirements for a web-based library management system.

#### 1.2 Scope

The scope of this SRS includes the user registration, book search, borrowing, returning, admin management, real-time availability tracking, reporting, and user session handling features.

#### 1.3 Definitions, Acronyms, and Abbreviations

- Book: A unique item in the library's inventory with title, author, ISBN, availability status.
- Admin User: A user account with elevated privileges to manage book records.
- User Session: A period of active use by a user.
- Inactivity: No user interaction for 30 minutes.
- Concurrent Users: Simultaneously logged-in users accessing the system.

#### 1.4 References

ANSI/IEEE STD 830-1998 - IEEE Standard for Software Requirements Specifications.

#### 1.5 Overview

The library management system provides an accessible and efficient means for users to register, search for books, borrow up to three books at a time, return books, and receive reminders when necessary. Admin users can manage book records, view real-time availability, and generate monthly reports.

### 2. Overall Description

#### 2.1 Product Perspective

The library management system is a web-based application accessible through modern browsers such as Chrome, Firefox, and Safari.

#### 2.2 Product Functions

- User Registration & Login (REQ-001)
- Book Search (REQ-002)
- Borrowing & Returning Books (REQ-003, REQ-004)
- Email Reminders (REQ-005)
- Admin Management (REQ-006)
- Real-Time Availability Tracking (REQ-007)

- Report Generation (REQ-008)
- User Session Handling (REQ-012)

## 2.3 User Characteristics

Users include general library patrons and administrators with varying technical proficiency levels.

## 2.4 Constraints

The system shall support at least 500 concurrent users without performance degradation (REQ-009).  
All user passwords must be stored encrypted (REQ-010).

## 2.5 Assumptions and Dependencies

This SRS assumes the availability of a reliable email service for sending reminders. The system's functionality depends on the integrity and completeness of book records provided by the administrators.

## 3. Functional Requirements

- REQ-001: The system shall allow users to register using their email and password.
- REQ-002: Users shall be able to search for books by title, author, or ISBN.
- REQ-003: Users should be able to borrow up to 3 books at a time.
- REQ-004: Borrowed books must be returned within 14 days.
- REQ-005: If a book is not returned on time, the system shall automatically send an email reminder to the user.
- REQ-006: Admin users should be able to add, update, and delete book records.
- REQ-007: The system shall show real-time availability of each book.
- REQ-008: The system shall generate monthly reports of borrowed and returned books for the admin.

## 4. Non-Functional Requirements

### 4.1 Performance Requirements

N/A (not applicable as performance requirements are not explicitly stated in the given requirements)

### 4.2 Security Requirements

All user passwords must be stored encrypted (REQ-010).

### 4.3 Usability Requirements

Not specified in the provided requirements.

### 4.4 Reliability Requirements

N/A (not applicable as reliability requirements are not explicitly stated in the given requirements)

### 4.5 Maintainability Requirements

Admin users should be able to add, update, and delete book records (REQ-006).

### 4.6 Portability Requirements

The system shall be accessible from modern web browsers including Chrome, Firefox, and Safari (REQ-011).

## 5. External Interface Requirements

### 5.1 User Interfaces

N/A (not applicable as user interface requirements are not explicitly stated in the given requirements)

### 5.2 Hardware Interfaces

Not specified in the provided requirements.

### 5.3 Software Interfaces

Not specified in the provided requirements.

### 5.4 Communication Interfaces

**The system shall communicate with a reliable email service for sending reminders (REQ-005). [AI-SUGGESTED]**

## 6. System Constraints

N/A (not applicable as system constraints are not explicitly stated in the given requirements)

## 7. Assumptions and Dependencies

This SRS assumes the availability of a reliable email service for sending reminders (REQ-005). The system's functionality depends on the integrity and completeness of book records provided by the administrators (REQ-006).